

ALAAELDIN ABDELAAL

Full Stack Developer

☎ 07776939174 @ alaaeldinabdelaal94@gmail.com 🔗 <https://www.linkedin.com/in/alaeldinsafwat/>

SUMMARY

Software Engineer with 2+ years of industry experience designing and building scalable backend systems and full stack applications. Strong focus on backend development using **Node.js**, **Go**, and **Python**, with production experience in authentication, authorization, and secure system design. Background in **Linux**, **embedded systems**, and system-level programming, enabling strong understanding of performance, reliability, and end-to-end architecture. Comfortable owning features from design through deployment in complex environments.

EXPERIENCE

Software Engineer

GE Vernova

📅 09/2024 - Present 📍 Rugby, United Kingdom

- Developed a **device authentication and authorization system** using **Node.js**, **Typescript**, **Angular**, and security protocols including **SSO**, **LDAP**, **RADIUS**, and **OAuth2**, with a frontend built in **Angular**, **HTML**, and **CSS**.
- Developed a custom **LDAP** server in **Go** and **JavaScript** responsible for user lifecycle management (creation, deletion, authentication).
- Built internal automation tools to update and manage HMI symbols, supporting both GUI and CLI workflows.
- Trained and integrated a **machine learning** model to predict ship draft using environmental data.
- Implemented **CI/CD** pipelines to automate releases, execute unit tests, and perform vulnerability scanning.
- Collaborated with cross-functional teams to deliver secure and reliable systems in production environments.
- Created an **automation tool** to update HMI symbols (aliases, positioning, and transformation handling), accessible via both **GUI** and **command line**.
- Worked with **React** for front-end development in small to medium features.

Software Engineer

ALTEN - Innovation Lab

📅 09/2023 - 09/2024 📍 Derby, United Kingdom

- Developed **RESTful** backend services using **Flask**, **SQLAlchemy**, and **PyTorch** for real-time **object detection** in mobile video streams.
- Built backend systems for 3D CAD model generation from mobile-captured images using **computer vision** and 3D reconstruction techniques.
- Applied **PyTorch** and **OpenCV** to production-oriented **ML** pipelines, integrating models into backend services.
- Worked in an **Agile** environment, contributing to system design, implementation, and testing.

Associate Lecturer

University of Central Lancashire

📅 10/2021 - 08/2023 📍 Preston, United Kingdom

- Taught **C#** and **C++** programming, focusing on clean code, maintainability, and problem-solving.
- Delivered modules on **embedded systems** and **robotics**, strengthening expertise in hardware-software integration.
- Mentored students in designing, debugging, and scaling complex systems.
- Developed strong communication and technical leadership skills applicable to collaborative engineering teams.

SKILLS

Node JS	Angular	HTML	CSS
RESTful APIs	Sequelize	CI/CD	
GO	SQLAlchemy	Linux	Docker
Python	C#	Machine Learning	
Deep Learning	GitHub	Agile	
Scrum	JIRA	MATLAB	GIT
C++	Embedded systems		
PostgreSQL	NumPy	pandas	

EDUCATION

MSc in Applied Data Science

University of Central Lancashire

📅 09/2021 - 12/2022

B.Eng. in Mechatronics and Intelligent Machines (1st Class)

University of Central Lancashire

📅 2020 - 2021

B.Eng. in Mechatronics

Future University in Egypt

📅 2016 - 2019

AWARDS

💎 **ECPC Computer programming contest**

💎 **3rd place in Egypt in the EAEAT National Competition for Industrial Robotics Applications**

💎 **IEEE Xtreme Competition ambassador**